Submission Document 2
Architectural Specification & End-to-End User Flow
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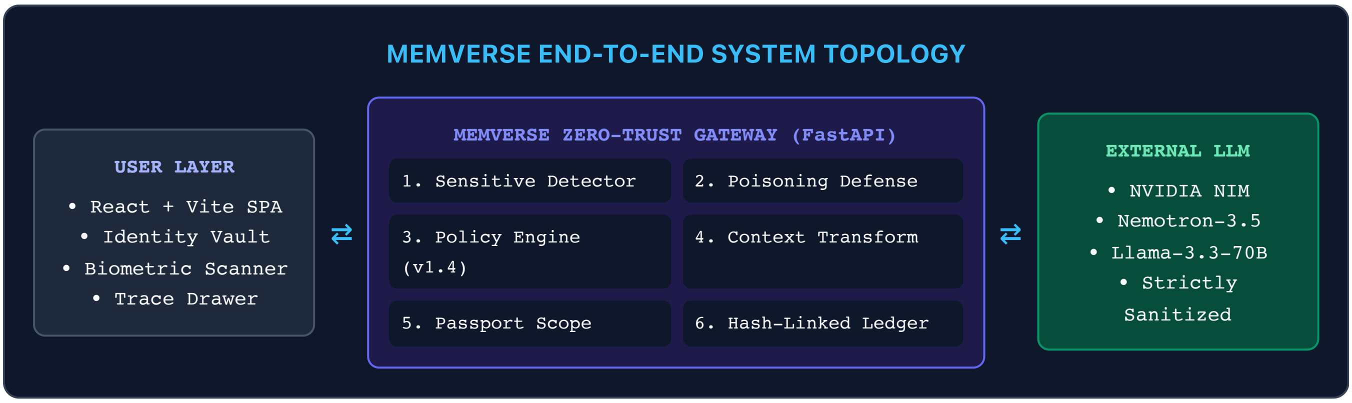
MEMVERSE Architecture & User Journey

Complete Technical Architecture, 12-Stage Pipeline, and Multimodal Data Flow

1. High-Level System Architecture

MEMVERSE is built upon a decoupled **Client & Gateway & LLM Engine** topology. The browser UI communicates strictly with the local/hosted MEMVERSE Gateway, which isolates private storage and manages external model egress.



2. The 12-Stage Zero-Trust Execution Pipeline

Every multimodal user interaction passes synchronously through 12 rigorous security stages inside the gateway:

1. **Request Capture & Ingestion:** Normalizes incoming prompt, user destination parameter, purpose tag, and session identifier.
2. **Sensitive Entity Detection (NER):** Scans for names, phone numbers, emails, government IDs, organization tokens, and financial amounts.
3. **Poisoning & Injection Defense:** Evaluates prompt against adversarial injection patterns, instruction overrides, and memory jailbreaks.
4. **Memory Candidate Retrieval:** Fetches stored memories matching user persona and verifies active cryptographic Passport state.
5. **Policy Engine Evaluation (v1.4 Matrix):** Determines privacy actions per detected field based on sensitivity (`HIGH` → SUPPRESS, `MEDIUM` → GENERALIZE, `LOW` → ALLOW).
6. **Context Transformation:** Executes rule-based replacements (e.g. *"Alex"* → *"person"*, *"24"* → *"18–24"*, *"Delhi"* → *"Northern India"*).
7. **Passport Scope Validation:** Ensures memory has not expired (TTL), has granted consent, and is bound strictly to the declared purpose.
8. **Approved Context Synthesis:** Assembles the clean context block and sanitized prompt. Raw data is strictly excluded.
9. **Egress Validation Gate:** Scans the final outgoing payload before network egress to verify zero prohibited sensitive tokens remain.

10. **External Model Execution:** Transmits sanitized context to NVIDIA NIM endpoint and streams response deltas back to the gateway.
11. **Response Ingress Gate:** Inspects external model completion to ensure no hallucinated memory leaks or policy violations occurred.
12. **Cryptographic Ledger Receipt:** Generates a SHA-256 hash-linked receipt linking previous event hash, request metadata, and policy version.

3. Multimodal Pipeline: Documents & Biometrics

MEMVERSE extends zero-trust protection beyond text to enterprise documents and facial biometric inputs:

📄 Document Ingestion (PDF / DOCX)

- **Structural Parsing:** Extracts tabular data, semester records, and textual contents via PyPDF / PDFPlumber.
- **Automated PII Masking:** Detects and redacts student roll numbers, signatures, and confidential marks before context generation.
- **Provenance Trace:** Links extracted semantic facts to verifiable memory passports.

📷 Biometric Photo Scanner

- **Client-Side Detection:** Runs face-api.js directly in the browser to isolate facial boundaries locally.
- **Granular Consent Prompt:** Displays extracted portrait and requests explicit user consent before storing identity attributes.
- **Metadata Stripping:** Removes EXIF coordinates, camera serials, and raw high-res biometric data prior to model inference.

4. Complete End-to-End User Flow

The interactive journey follows a transparent, inspectable sequence:

Stage	User Action	MEMVERSE Gateway Operation	Auditable Output
1. Connect	Enters app or uploads PDF / image	Generates cryptographic identity passport; stores encrypted memory locally	Passport with UUID & SHA-256 hash
2. Query	Submits prompt in ChatView	Passes query through 12-stage pipeline; transforms raw PII to generalized bands	Sanitized system prompt
3. Stream	Views real-time streamed answer	Connects to NVIDIA NIM via SSE; streams response to client	Tailored career/academic synthesis
4. Inspect	Opens *Inspect Trace* drawer	Displays radar visualizer, entity logs, stage timings, and diff viewer	Interactive 12-stage visual trace
5. Audit	Checks Event Ledger tab	Verifies cryptographic proof and Merkle-style hash link integrity	Verifiable Receipt with tamper check

5. Security Guarantees & Verification Summary

🔒 Verifiable Security Guarantees:

1. **No Direct Model Access:** The browser client has *zero* connection to NVIDIA API keys or model endpoints.
2. **Fail-Closed Privacy:** If any stage encounters an invalid token, missing consent, or unverified passport, egress is immediately blocked.
3. **Non-Repudiation:** The hash-linked ledger mathematically proves what exact data was transformed and sent at any timestamp.